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Random matrix theory and the Riemann zeros I: three- and four-point correlations

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Abstract. The non-trivial zeros of the Riemann zeta-function have been conjectured to be pairwise distributed like the eigenvalues of matrices in the Gaussian Unitary Ensemble (GUE) of random matrix theory. They therefore behave like the energy levels of quantum systems whose classical limit is strongly chaotic and non-invariant with respect to time-reversal. We show that this analogy extends directly to higher-order statistics. Starting with an explicit formula relating the zeros to the prime numbers (the analogue of the Gutzwiller trace formula of quantum chaology), we demonstrate that the 3-point and 4-point zero correlation functions are asymptotically equivalent to the corresponding GUE results. Our method centres around a Hardy–Littlewood conjecture concerning the distribution of the primes. The calculation generalises a previous study of 2-point correlations and involves the introduction of several new techniques. These will form the basis of a demonstration, to be described in another paper, that the equivalence extends to the general n-point correlation function.

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1. Introduction

Random matrix theory predicts specific correlations between the eigenvalues of matrices whose elements are picked at random from various symmetry-invariant ensembles (Mehta 1991). This has proved to be of considerable importance in quantum chaology, because extensive numerical investigations support the conjecture that, in the semiclassical limit, the same correlations generically characterize the energy-level spectra of systems whose classical motion is chaotic (Bohigas *et al* 1984a, Berry 1987). Given that in this limit the quantum energy levels E_n are related to classical periodic orbits via Gutzwiller's trace formula (Gutzwiller 1971), this naturally leads to the question of identifying those properties of the orbits that determine the spectral statistics, and the semiclassical mechanism by which they do so.

The relationship between eigenvalue correlations and periodic orbit distributions was first considered by Hannay and Ozorio de Almeida (1984) and Berry (1985). Using a sum rule relating the proliferation of periodic orbits to their stabilities in ergodic systems, they confirmed that the long-range (large $|x_1 - x_2|$) limit of the 2-point autocorrelation function $R_2(x_1, x_2)$, defined by

$$R_2(x_1, x_2) = \langle d(E + x_1)d(E + x_2)\rangle_E \tag{1}$$

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where d is the density of states

$$d(E) = \sum_n \delta(E - E_n) \tag{2}$$

and the angular brackets denote an average with respect to the energy E, coincides with the random matrix conjecture. To extend this semiclassical approach to all values of $x_1 - x_2$ would, however, require much more detailed information about the correlations between the actions of different periodic orbits than is currently available.

A guide as to what this information might ultimately turn out to be may be obtained by inverting the problem; that is, by asking what form the action correlations must take for them to be consistent with the random matrix theory expressions for R_2 (Keating 1993). Unfortunately, even though the results are simple to derive and are to some extent confirmed by numerical tests (Argaman *et al* 1993), it has not proved possible to obtain them independently of the random matrix conjecture; nor does there appear to be any sign of how they might emerge from an analysis based solely on classical mechanics. Consequently, it must be concluded that there has been no real progress, beyond the work of Hannay and Ozorio de Almeida (1984) and Berry (1985), in understanding semiclassically how the full random matrix form of R_2 might arise for any given dynamical system, or class of such systems. (The 2-point correlation function has been calculated for certain models, but these are non-generic in the sense that they possess special number-theoretical symmetries which conspire largely to destroy the spectral correlations (Keating 1991a, Bogomolny *et al* 1994)).

In the light of this problem, it is natural to seek mathematical examples that can be used as a guide to how random matrix statistics arise in a spectrum, and to the techniques required to derive them, via a trace-type formula, from periodic orbit correlations. Such an example would clearly have to satisfy the criteria that, first, the analogue of the spectral statistics satisfies the random matrix conjecture, and second, that the analogues of the periodic orbit correlations are sufficiently well understood for them to be used as the basis of a systematic analysis. Remarkably, such an example is provided by the Riemann zeta-function.

The connections between the theory of the Riemann zeta-function and that of quantum chaology, first highlighted by Berry (1985, 1986), are reviewed in Keating (1993). For our purposes, the essential results are that all of the criteria mentioned above are satisfied. First, numerical evidence (Odlyzko 1989) supports the conjecture (Montgomery 1973) that the 2-point autocorrelation function, defined as in (1) and (2), of the imaginary parts E_n of the complex zeros of the zeta function is asymptotically (as $E \to \infty$) equivalent to the corresponding Gaussian Unitary Ensemble (GUE) random matrix result:

$$R_2(x_1, x_2) = \bar{d}^2 - \frac{\sin^2\left[(x_1 - x_2)\pi\bar{d}\right]}{\pi^2(x_1 - x_2)^2} \tag{3}$$

where $\bar{d} = \langle d \rangle_E$ is the mean density of states. Second, there is an explicit formula for the density of states of the zeros, defined as in (2), that is directly analogous to the Gutzwiller trace formula for a completely chaotic system:

$$d(E) = \bar{d}(E) - \frac{1}{\pi} \sum_p \sum_{n=1}^{\infty} \frac{\ln p}{p^{n/2}} \cos(En \ln p) \tag{4}$$

where the first sum includes all primes p, and the mean density of states is given by

$$\bar{d}(E) = \frac{1}{2\pi} \left\{ \mathrm{Re} \frac{\Gamma'(\frac{1}{4} - \mathrm{i}\frac{E}{2})}{\Gamma(\frac{1}{4} - \mathrm{i}\frac{E}{2})} - \ln \pi \right\} \tag{5}$$

or, using Stirling's formula, as $E \to \infty$ by

$$\bar{d}(E) \sim \frac{1}{2\pi} \ln E. \tag{6}$$

Third, the correlations between the primes (whose logarithms play the role of the periods of the classical orbits) have been much studied and are the subject of a series of celebrated conjectures due to Hardy and Littlewood (1923). In fact, it is known that the 2-point correlation function result (3) is essentially equivalent to a prime correlation formula that follows from one of the Hardy–Littlewood conjectures (Goldston and Montgomery 1984, Keating 1993, Argaman *et al* 1993).

Our aim in the present work is to pursue the analogy provided by the Riemann zeta-function to its natural conclusions in order to derive as much information as possible about the semiclassical origins of GUE statistics. Specifically, our intention is to generalize the previous study of 2-point statistics to the n-point correlation function $R_n(x_1, x_2, \ldots, x_n)$. The approach to be taken is a direct extension of the method for calculating 2-point correlation information developed in Keating (1991b) and Keating (1993), and recently used to establish similar results for other L-functions in Bogomolny and Lebœuf (1994). It is based on a heuristic application of one of the Hardy–Littlewood conjectures to evaluate a formula, derived using (4), which relates R_n to an n-fold sum over the prime numbers. Our main result is that the random matrix conjecture holds not just when $n = 2$, but for all $n \geqslant 2$.

The calculation will be split into two papers. In this, the first, we shall use the explicit computation of R_3 and R_4 to identify and evaluate the most important prime correlation contributions. This will require the introduction of several new techniques. The combinatorial methods needed to incorporate these into the corresponding analysis of the general formula for R_n and to show that the result is equivalent to the appropriate GUE expression will be given in a second paper.

Statistical measures which include the three- and four-point correlations between the Riemann zeros have been studied numerically by Odlyzko (1987). His results for the skewness γ_1 and the excess γ_2 show close agreement with the corresponding GUE expressions. Unfortunately, both of these quantities are dominated by the lower-order correlations and so it is not clear to what extent the data support our findings that the *irreducible* three- and four-point correlation functions (defined by equations (21) and (34)) individually coincide with the GUE results. The same is also true for physical systems: γ_1 and γ_2 were computed numerically for the stadium billiard by Bohigas *et al* (1984b) and were obtained from an ensemble of nuclear data in Bohigas *et al* (1985), but the irreducible correlation functions themselves have not been studied. Statistics which include contributions from R_n with $n > 4$ have, to the best of our knowledge, not previously been investigated.

It is, of course, known that on scales that are asymptotically much larger than the mean spacing, there are deviations from the random matrix conjecture for R_2, both for semiclassical spectra (Berry 1985) and for the zeros of the Riemann zeta-function (Berry 1988). These are also present in the higher-order correlations, but will be ignored in the present work, whose main goal is, as stated above, to identify the features that give rise to generic, random matrix behaviour at smaller distances. Specifically, our results apply for unfolded correlation lengths $y_i = x_i \bar{d}(E)$ that satisfy $y_i / \ln E \to 0$ as $E \to \infty$; thus they apply to the usual case in which the limit $E \to \infty$ is taken with all the y_i fixed (i.e. under the conditions for which Montgomery's conjecture holds). The present analysis can be extended to the non-universal range in which $y_i / \ln E \gg 1$ exactly as in Berry (1988).

Finally, we note that at a late stage in the preparation of this manuscript we learned of complementary calculations for R_3 by Hejhal (1994), and for R_n by Rudnick and Sarnak (1994). Their approach, based on what is commonly termed the diagonal approximation, is restricted to particular ranges of values of the correlation variables x_n, but is rigorous, whereas ours is unrestricted but heuristic in that it makes essential use of the unproved Hardy–Littlewood conjecture and involves further uncontrolled approximations. In the regions of overlap, the results agree with each other.

2. Prime correlations

Our calculation of the 3-point and 4-point correlations between the complex zeros of the Riemann zeta-function will be based on the following conjecture of Hardy and Littlewood (1923) (conjecture D in the notation of their paper). Let (m, n) denote the highest common factor of the integers m and n. When $(m, n) = 1$, define $\Pi(m, n, k, N)$ to be the number of pairs of primes (p, q) satisfying

$$mp = nq + k \tag{7}$$

and $p < N$, where k is a positive integer. Then if $(k, m) = (k, n) = 1$ and just one of k, m, n is even,

$$\Pi(m, n, k, N)\frac{\ln^2 N}{N} \sim \frac{2C_2}{n} \prod_{\substack{p\geqslant 3 \\ p|k \\ p\nmid m \\ p\nmid n}} \frac{p-1}{p-2} \tag{8}$$

as $N \to \infty$, where the product includes all odd prime divisors of k, m and n, and C_2 is a constant given by

$$C_2 = \prod_{q\geqslant 3}\left(1 - \frac{1}{(q-1)^2}\right) = 0.66016\ldots, \tag{9}$$

the product running over all odd primes q; otherwise

$$\Pi(m, n, k, N)\frac{\ln^2 N}{N} \to 0 \tag{10}$$

as $N \to \infty$.

In the case when $m = n = 1$, a probabilistic (and hence heuristic) justification of this conjecture was given by Lord Cherwell (1946) and in Keating (1993), and the general result stated above may be motivated by a simple extension of these arguments. A useful interpretation, directly related to this approach, follows from the observation that $N/\ln N$ is asymptotically the number of primes less than N, and so for integers k, m and n satisfying the above criteria the probability that, given a large prime p, $q = (mp - k)/n$ is also a prime (and hence an integer) is

$$P(m, n, k) = \frac{2C_2}{n \ln q} \prod_{\substack{p\geqslant 3 \\ p|k \\ p\nmid m \\ p\nmid n}} \frac{p-1}{p-2} \tag{11}$$

and is zero otherwise.

In the course of our analysis, we shall frequently have cause to evaluate expressions of the form

$$F(m, n, E) = \sum_{k \neq 0} \exp\left(\frac{iEk}{\alpha}\right) P(m, n, k) \tag{12}$$

in the limit as $E \to \infty$, where k takes all positive and negative integer values other than 0. As shown in Keating (1993), a good approximation to such sums follows from replacing P by its local average over k:

$$\bar{P}(m, n, k) = \frac{1}{k} \sum_{k'=1}^{k} P(m, n, k'). \tag{13}$$

In the case when $m = n = 1$, this average was calculated to be

$$\bar{P}(1, 1, k) \approx \frac{1}{\ln q}\left(1 - \frac{1}{2k}\right) \tag{14}$$

when $k \gg 1$. (We note in passing that if the primes were pairwise randomly (Poisson) distributed, this would correspond to keeping only the first term in the bracket in (14).) It was then demonstrated that in the limit $E \to \infty$, F may be approximated by substituting (14) into (12), replacing the sum by an integral, and using the evenness of P with respect to k to give

$$F(1, 1, E) \approx -\frac{1}{\ln q} \int_{k_0 E/\alpha}^{\infty} \cos x \frac{dx}{x} \tag{15}$$

where k_0 is a small constant. Only the second term in the bracket in (14) contributes to this result; that is, the correction to randomness in the pairwise distribution of the primes dominates, even though it is small.

Clearly if $\alpha = o(E)$ (i.e. if $E/\alpha \to \infty$), the integral in (15) vanishes as $E \to \infty$. If, however, $\alpha^{-1} = o(E^{-1})$ (i.e. $E/\alpha \to 0$), then a leading-order asymptotic evaluation gives $\ln(\alpha/E)$. These two limiting cases may be combined to yield the uniform approximation

$$F(1, 1, E) \approx \frac{1}{\ln q} \ln\left(\frac{E}{\alpha}\right) \Theta\left(\ln\left(\frac{\alpha}{E}\right)\right) \tag{16}$$

where Θ is the unit step function. This result represents one of the basic elements of our analysis.

The approximation (16) is essentially all that is needed to evaluate the 2-point correlation function R_2, or, equivalently, its Fourier transform, the spectral form factor (Keating 1991b, 1993). In progressing to the higher-order statistics, we shall need to consider the more general case corresponding to when m and n are both odd and their prime factors can be considered large. The main difficulty that this throws up is the incorporation of the coprimality conditions on k stated before equation (8) into the sum (12). We deal with this using an inclusion/exclusion expansion; that is, by first ignoring the restrictions on k, and then by systematically re-imposing them through the subtraction of a series of terms that isolate the disallowed values.

To derive the general form of this expansion, we start with

$$F(m,n,E) = \sum_{\substack{k\neq 0 \\ (m,n,k)=1}} \exp\left(\frac{\mathrm{i}Ek}{\alpha}\right) \frac{2C_2}{n\ln q} \prod_{\substack{p\geqslant 3 \\ p\nmid k \\ p\mid m \\ p\mid n}} \frac{p-1}{p-2} \tag{17}$$

where the sum includes all even integers k (except for $k = 0$) that are coprime to both m and n. The key step is to write this in the form

$$\begin{aligned} F(m,n,E) = \frac{1}{n\ln q}\Bigg[& \sum_{k\neq 0} \exp\left(\frac{\mathrm{i}Ek}{\alpha}\right) 2C_2 \prod_{\substack{p\geqslant 3 \\ p\nmid k \\ p\mid m \\ p\mid n}} \frac{p-1}{p-2} \\ & - \sum_{p_1\mid mn} \sum_{k\neq 0} \exp\left(\frac{\mathrm{i}Ekp_1}{\alpha}\right) 2C_2 \prod_{\substack{p\geqslant 3 \\ p\nmid k \\ p\mid m \\ p\mid n}} \frac{p-1}{p-2} \\ & + \sum_{p_1\mid mn} \sum_{p_2\mid mn} \sum_{k\neq 0} \exp\left(\frac{\mathrm{i}Ekp_1p_2}{\alpha}\right) 2C_2 \prod_{\substack{p\geqslant 3 \\ p\nmid k \\ p\mid m \\ p\mid n}} \frac{p-1}{p-2} \\ & - \sum_{p_1\mid mn} \sum_{p_2\mid mn} \sum_{p_3\mid mn} \sum_{k\neq 0} \exp\left(\frac{\mathrm{i}Ekp_1p_2p_3}{\alpha}\right) 2C_2 \prod_{\substack{p\geqslant 3 \\ p\nmid k \\ p\mid m \\ p\mid n}} \frac{p-1}{p-2} + \ldots \Bigg] \end{aligned} \tag{18}$$

where now the k-sums are no longer restricted to integers that are coprime to m and n, and the additional sums include all distinct prime divisors P_i ($\neq p_j \neq p_k$ etc) of mn. Since these prime factors are assumed large (and finite in number), each term in (18) may be approximated by making the replacement

$$2C_2 \prod_{\substack{p\geqslant 3 \\ p\nmid k \\ p\mid m \\ p\mid n}} \frac{p-1}{p-2} \to 1 - \frac{1}{2k} \tag{19}$$

as in the case when $m = n = 1$. The k-sums may then be approximated as above, giving the generalization of (16) to be

$$\begin{aligned} F(m,n,E) \approx \frac{1}{n\ln q}\Bigg[& \ln\left(\frac{E}{\alpha}\right) \Theta\left(\ln\left(\frac{\alpha}{E}\right)\right) - \sum_{p_1\mid mn} \ln\left(\frac{Ep_1}{\alpha}\right) \Theta\left(\ln\left(\frac{\alpha}{Ep_1}\right)\right) \\ & + \sum_{p_1\mid mn} \sum_{p_2\mid mn} \ln\left(\frac{Ep_1p_2}{\alpha}\right) \Theta\left(\ln\left(\frac{\alpha}{Ep_1p_2}\right)\right) \\ & - \sum_{p_1\mid mn} \sum_{p_2\mid mn} \sum_{p_3\mid mn} \ln\left(\frac{Ep_1p_2p_3}{\alpha}\right) \Theta\left(\ln\left(\frac{\alpha}{Ep_1p_2p_3}\right)\right) + \ldots \Bigg] \end{aligned} \tag{20}$$

as $E \to \infty$.

The approximation (20) represents the main result of this section. Just as (16) formed the basis of the calculation of the 2-point statistics of the Riemann zeros (Keating 1991b, 1993), its extension will be used in the next two sections to compute the 3-point and 4-point correlations, and in a subsequent paper to evaluate the general n-point statistic.

3. The 3-point correlation function

Taking the non-trivial zeros of the Riemann zeta-function $\zeta(s)$ to lie at $s = 1/2 + iE_n$ and assuming the correctness of the Riemann hypothesis that $\mathrm{Im}\, E_n = 0\ \forall n$, a density function may be defined exactly as in (2). If this is split into its mean $\bar{d}$, given by (5) and asymptotically by (6), and fluctuations $\tilde{d}$ thereabout, the irreducible 3-point autocorrelation function is defined by

$$\tilde{R}_3(x_1, x_2, x_3) = \langle \tilde{d}(E + x_1)\tilde{d}(E + x_2)\tilde{d}(E + x_3)\rangle_E \tag{21}$$

where the angular brackets denote a local average with respect to E, that is, an average over a range ΔE that satisfies $\Delta E \bar{d} \to \infty$ as $E \to \infty$; for example, a suitable candidate is to take the size of the averaging range to be E itself, i.e. to define

$$\langle f(E)\rangle_E = \frac{1}{E}\int_0^E f(E')\,\mathrm{d}E'. \tag{22}$$

The GUE random matrix result for $\tilde{R}_3$ is that as $E \to \infty$

$$\tilde{R}_3(x_1, x_2, x_3) = \begin{vmatrix} 0 & s(x_1, x_2) & s(x_1, x_3) \\ s(x_2, x_1) & 0 & s(x_2, x_3) \\ s(x_3, x_1) & s(x_3, x_2) & 0 \end{vmatrix} \tag{23}$$

where

$$s(x_i, x_j) = \frac{\sin[(x_i - x_j)\pi\bar{d}(E)]}{\pi(x_i - x_j)}; \tag{24}$$

that is, if we scale out the mean density by defining $y_i = x_i\bar{d}(E)$ and divide $\tilde{R}_3$ by $\bar{d}^3$, the result tends to a well defined limit.

Expanding the determinant in (23) gives

$$\tilde{R}_3(x_1, x_2, x_3) = -\frac{1}{2\pi^3(x_1 - x_2)(x_2 - x_3)(x_3 - x_1)}\{\sin[2\pi(x_1 - x_2)\tilde{d}] + \sin[2\pi(x_2 - x_3)\tilde{d}] + \sin[2\pi(x_3 - x_1)\tilde{d}]\} \tag{25}$$

and it is this form that will be shown to hold for the Riemann zeros. To do so, we shall use for $\tilde{d}$ its representation as a sum over primes as given by (4) (recall that by definition $\tilde{d} = d - \bar{d}$). Substituting this expression into (21) then provides an explicit formula for $\tilde{R}_3$. Following the approximations motivated in Keating (1991b) and Keating (1993) (which

show that in this formula the contribution from the sum over the integers n in (4) can, to leading order, be approximated simply by retaining the $n = 1$ term) then leads to

$$\begin{aligned}\tilde{R}_3(x_1, x_2, x_3) \approx -\frac{1}{4\pi^3}\,\mathrm{Re}\sum_{p_1}\sum_{p_2}\sum_{p_3}\frac{\ln p_1 \ln p_2 \ln p_3}{\sqrt{p_1 p_2 p_3}}\\ \times \langle \exp(\mathrm{i}E\ln(p_1p_2p_3) + \mathrm{i}x_1\ln p_1 + \mathrm{i}x_2\ln p_2 + \mathrm{i}x_3\ln p_3)\\ + \exp(\mathrm{i}E\ln(p_1/p_2p_3) + \mathrm{i}x_1\ln p_1 - \mathrm{i}x_2\ln p_2 - \mathrm{i}x_3\ln p_3)\\ + \exp(\mathrm{i}E\ln(p_2/p_1p_3) + \mathrm{i}x_2\ln p_2 - \mathrm{i}x_1\ln p_1 - \mathrm{i}x_3\ln p_3)\\ + \exp(\mathrm{i}E\ln(p_3/p_2p_1) + \mathrm{i}x_3\ln p_3 - \mathrm{i}x_2\ln p_2 - \mathrm{i}x_1\ln p_1)\rangle_E\,. \end{aligned} \tag{26}$$

Because of the E-average, the first term in the angular brackets makes no contribution. Also, the third and fourth terms may be obtained directly from the second via $x_1 \leftrightarrow x_2$ and $x_1 \leftrightarrow x_3$, respectively. Hence it suffices to consider just the contribution from the second term, which we denote by

$$\begin{aligned}\tilde{r}_3(x_1, x_2, x_3) \approx -\frac{1}{4\pi^3}\,\mathrm{Re}\sum_{p_1}\sum_{p_2}\sum_{p_3}\frac{\ln p_1 \ln p_2 \ln p_3}{\sqrt{p_1 p_2 p_3}}\\ \times \langle \exp(\mathrm{i}E\ln(p_1/p_2p_3) + \mathrm{i}x_1\ln p_1 - \mathrm{i}x_2\ln p_2 - \mathrm{i}x_3\ln p_3)\rangle_E\,. \end{aligned} \tag{27}$$

The effect of the E-average in (27) is that the main contributions to the triple prime sum come from terms that satisfy $p_1 \approx p_2p_3$. We are thus led to consider the following problem: given primes p_2 and p_3, what is the probability that p_2p_3+k, where $k \ll p_2p_3$ is an integer satisfying $(k, p_2) = (k, p_3) = 1$, is also a prime? Once this is known, the sum over p_1 can be replaced by a sum over k, with the approximation

$$\exp\left(\mathrm{i}E\ln\left(\frac{p_1}{p_2p_3}\right)\right) \to \exp\left(\frac{\mathrm{i}Ek}{p_2p_3}\right). \tag{28}$$

It will be argued below that the primes p_2 and p_3 can both be considered to be large.

The probability in question is obviously a particular example of the general case covered by the Hardy–Littlewood conjecture and reviewed in section 2. The resulting sum over k is thus precisely of the form (12) and hence can be evaluated using (20), the restriction that k be coprime to both p_2 and p_3 being incorporated by an inclusion/exclusion expansion with respect to these primes. The result is that

$$\begin{aligned}\tilde{r}_3(x_1, x_2, x_3) \approx -\frac{1}{4\pi^3}\,\mathrm{Re}\sum_{p_2}\sum_{p_3}\frac{\ln p_2 \ln p_3}{p_2p_3}\exp[\mathrm{i}(x_1-x_2)\ln p_2 + \mathrm{i}(x_1-x_3)\ln p_3]\\ \times\left[\ln\left(\frac{E}{p_2p_3}\right)\Theta\left(\ln\left(\frac{p_2p_3}{E}\right)\right) - \ln\left(\frac{E}{p_2}\right)\Theta\left(\ln\left(\frac{p_2}{E}\right)\right)\right.\\ \left. - \ln\left(\frac{E}{p_3}\right)\Theta\left(\ln\left(\frac{p_3}{E}\right)\right)\right] \end{aligned} \tag{29}$$

where we have used the fact that the final term in the inclusion/exclusion expansion vanishes under the E-average in (27).

The prime sums in (29) may be approximated using

$$\sum_p g(p) \approx \int_1^\infty \frac{g(x)}{\ln x}\,\mathrm{d}x, \tag{30}$$

the justification for which follows from the prime number theorem provided the dominant contributions come from the large primes. Setting the lower limit equal to 1 is appropriate if the integral converges there when $E \to \infty$, as it does in our case (Keating 1993). The result is that

$$\tilde{r}_3(x_1, x_2, x_3) \approx -\frac{1}{4\pi^3} \operatorname{Re} \int_0^\infty \int_0^\infty \exp[\mathrm{i}(x_1 - x_2)\alpha_2 + \mathrm{i}(x_1 - x_3)\alpha_3] \times [(\ln E - \alpha_2 - \alpha_3)\Theta(\alpha_2 + \alpha_3 - \ln E) - (\ln E - \alpha_2)\Theta(\alpha_2 - \ln E) - (\ln E - \alpha_3)\Theta(\alpha_3 - \ln E)]\, \mathrm{d}\alpha_2\, \mathrm{d}\alpha_3. \tag{31}$$

As $E \to \infty$, the double integral is dominated by the values of α_1 and α_2 corresponding to asymptotically large primes p_1 and p_2, confirming the assertion made above. This feature is common to all subsequent calculations and justifies the fact that $F(m, n, E)$ was calculated in section 2 for the case when the prime factors of m and n could be considered large.

A direct evaluation of the integral in (31) gives

$$\tilde{r}_3(x_1, x_2, x_3) \approx -\frac{1}{4\pi^3} \frac{\sin[(x_1 - x_2)\ln E] + \sin[(x_3 - x_1)\ln E]}{(x_1 - x_2)(x_2 - x_3)(x_3 - x_1)}. \tag{32}$$

Adding on the terms corresponding to $x_1 \leftrightarrow x_2$ and $x_1 \leftrightarrow x_3$ then leads to the final result for $\tilde{R}_3$:

$$\tilde{R}_3(x_1, x_2, x_3) \approx -\frac{1}{2\pi^3(x_1 - x_2)(x_2 - x_3)(x_3 - x_1)} \{\sin[(x_1 - x_2)\ln E] + \sin[(x_2 - x_3)\ln E] + \sin[(x_3 - x_1)\ln E]\}. \tag{33}$$

This is obviously equal to (25) when $\tilde{d}$ is given by (6), confirming that the three-point correlation function for the Riemann zeros, calculated using the Hardy–Littlewood conjecture, indeed corresponds to the random matrix GUE result.

4. The four-point correlation function

The irreducible four-point autocorrelation function is defined by

$$\tilde{R}_4(x_1, x_2, x_3, x_4) = \left\langle \tilde{d}(E + x_1)\tilde{d}(E + x_2)\tilde{d}(E + x_3)\tilde{d}(E + x_4) \right\rangle_E. \tag{34}$$

The GUE random matrix result is that when $E \to \infty$,

$$\tilde{R}_4(x_1, x_2, x_3, x_4) = \begin{vmatrix} 0 & s(x_1, x_2) & s(x_1, x_3) & s(x_1, x_4) \\ s(x_2, x_1) & 0 & s(x_2, x_3) & s(x_2, x_4) \\ s(x_3, x_1) & s(x_3, x_2) & 0 & s(x_3, x_4) \\ s(x_4, x_1) & s(x_4, x_2) & s(x_4, x_3) & 0 \end{vmatrix} \tag{35}$$

with s defined by (24).

Our aim, as before, is to show that the GUE result holds for the Riemann zeros by substituting for $\tilde{d}$ in (34) its representation as a sum over primes. Doing so gives that

$$\tilde{R}_4(x_1, x_2, x_3, x_4) = \tilde{r}_4^{(1)}(x_1, x_2, x_3, x_4) + \tilde{r}_4^{(1)}(x_2, x_1, x_3, x_4) + \tilde{r}_4^{(1)}(x_3, x_2, x_1, x_4) + \tilde{r}_4^{(1)}(x_4, x_2, x_3, x_1) + \tilde{r}_4^{(2)}(x_1, x_2, x_3, x_4) + \tilde{r}_4^{(2)}(x_3, x_2, x_1, x_4) + \tilde{r}_4^{(2)}(x_4, x_2, x_3, x_1) \tag{36}$$

with

$$\tilde{r}_4^{(1)}(x_1,x_2,x_3,x_4)=\frac{1}{8\pi^4}\,\mathrm{Re}\sum_{p_1}\sum_{p_2}\sum_{p_3}\sum_{p_4}\frac{\ln p_1\ln p_2\ln p_3\ln p_4}{\sqrt{p_1p_2p_3p_4}}$$
$$\times\left\langle\exp\left[\mathrm{i}E\ln\left(\frac{p_1}{p_2p_3p_4}\right)+\mathrm{i}(x_1\ln p_1-\mathrm{i}x_2\ln p_2-\mathrm{i}x_3\ln p_3-\mathrm{i}x_4\ln p_4)\right]\right\rangle_E \tag{37}$$

and

$$\tilde{r}_4^{(2)}(x_1,x_2,x_3,x_4)=\frac{1}{8\pi^4}\,\mathrm{Re}\sum_{p_1}\sum_{p_2}\sum_{p_3}\sum_{p_4}\frac{\ln p_1\ln p_2\ln p_3\ln p_4}{\sqrt{p_1p_2p_3p_4}}$$
$$\times\left\langle\exp\left[\mathrm{i}E\ln\left(\frac{p_1p_2}{p_3p_4}\right)+\mathrm{i}(x_1\ln p_1+\mathrm{i}x_2\ln p_2\right.\right.$$
$$\left.\left.-\mathrm{i}x_3\ln p_3-\mathrm{i}x_4\ln p_4)\right]\right\rangle_E. \tag{38}$$

In deriving this expression, we have used the fact that the E-average removes an additional term with $E\ln(p_1p_2p_3p_4)$ in its exponent, and the property that the dominant asymptotics is determined by retaining only the $n=1$ contribution to (4). We now proceed to calculate $r_4^{(1)}$ and $r_4^{(2)}$ using the expansion developed in section 2. This will involve the introduction of several new combinatorial techniques, which will be discussed as the necessity for them arises.

4.1. Calculation of $\tilde{r}_4^{(1)}$

As a result of the E-average in (37), the main contribution to $\tilde{r}_4^{(1)}$ comes from terms in the quadruple prime-sum that satisfy $p_1\approx p_2p_3p_4$. We are thus led to consider the situations when $p_2p_3p_4+k$ is a prime, with $k\ll p_2p_3p_4$ an integer. Obviously k must also satisfy the conditions $(k,p_2)=(k,p_3)=(k,p_4)=1$. Our aim is again to replace the p_1-sum in (37) by a sum over k, with the coprimality restrictions incorporated via an inclusion/exclusion expansion with respect to p_2, p_3 and p_4. Making the replacement

$$\exp\left[\mathrm{i}E\ln\left(\frac{p_1}{p_2p_3p_4}\right)\right]\to\exp\left[\frac{\mathrm{i}Ek}{p_2p_3p_4}\right] \tag{39}$$

then reduces the new sum to the form considered in section 2 and allows us to apply the general result (20) directly. This gives

$$\tilde{r}_4^{(1)}(x_1,x_2,x_3,x_4)\approx\frac{1}{8\pi^4}\int_0^\infty\int_0^\infty\int_0^\infty\exp[\mathrm{i}(x_1-x_2)\alpha_2+\mathrm{i}(x_1-x_3)\alpha_3+\mathrm{i}(x_1-x_4)\alpha_4]$$
$$\times G(\alpha_2,\alpha_3,\alpha_4,E)\,\mathrm{d}\alpha_2\mathrm{d}\alpha_3\mathrm{d}\alpha_4 \tag{40}$$

where

$$G(\alpha_2,\alpha_3,\alpha_4,E)=(\ln E-\alpha_2-\alpha_3-\alpha_4)\Theta(\alpha_2+\alpha_3+\alpha_4-\ln E)$$
$$-(\ln E-\alpha_2-\alpha_3)\Theta(\alpha_2+\alpha_3-\ln E)-(\ln E-\alpha_2-\alpha_4)\Theta(\alpha_2+\alpha_4-\ln E)$$
$$-(\ln E-\alpha_3-\alpha_4)\Theta(\alpha_3+\alpha_4-\ln E)+(\ln E-\alpha_2)\Theta(\alpha_2-\ln E)$$
$$+(\ln E-\alpha_3)\Theta(\alpha_3-\ln E)+(\ln E-\alpha_4)\Theta(\alpha_4-\ln E) \tag{41}$$

and where the sums over p_2, p_3 and p_4 have been approximated according to the prescription (30).

Evaluating the integrals in (40), we find

$$\tilde{r}_4^{(1)}(x_1, x_2, x_3, x_4) \approx -\frac{1}{8\pi^4}\left[\frac{\cos[(x_1-x_2)\ln E]}{(x_1-x_3)(x_2-x_3)(x_1-x_4)(x_2-x_4)} + \frac{\cos[(x_1-x_3)\ln E]}{(x_1-x_2)(x_3-x_2)(x_1-x_4)(x_3-x_4)} + \frac{\cos[(x_1-x_4)\ln E]}{(x_1-x_2)(x_4-x_2)(x_1-x_3)(x_4-x_3)}\right] \tag{42}$$

Adding together all of the $\tilde{r}_4^{(1)}$ contributions appearing in (36) then gives

$$\tilde{r}_4^{(1)}(x_1, x_2, x_3, x_4) + \tilde{r}_4^{(1)}(x_2, x_1, x_3, x_4) + \tilde{r}_4^{(1)}(x_3, x_2, x_1, x_4) + \tilde{r}_4^{(1)}(x_4, x_2, x_3, x_1) \approx -\frac{1}{4\pi^4}\sum_{i=1}^{4}\sum_{j>i}\frac{\cos[(x_i-x_j)\ln E]}{\prod_{k\neq i\atop k\neq j}(x_i-x_k)(x_j-x_k)}. \tag{43}$$

4.2. *Calculation of* $\tilde{r}_4^{(2)}$

For $r_4^{(2)}$, as defined by equation (38), the E-average implies that the most important terms in the quadruple prime-sum are those satisfying $p_1p_2 \approx p_3p_4$. There are four distinct ways in which this can be arranged: the first is that, for example, $p_1 = p_3$ and $p_2 = p_4$ (diagonal contributions); the second is that $p_1p_2 = p_3p_4 + k$ (type-I off-diagonal contributions); the third is that, for example, $p_1 = p_3$ and $p_2 = p_4 + k$ (semidiagonal contributions); and the fourth is that $mp_1 = np_3 + k$ whilst $np_2 = mp_4 + k'$ (type-II off-diagonal contributions). The first three cases can be dealt with directly using the methods already employed. The fourth requires new techniques which go beyond those previously developed.

4.2.1. Diagonal contributions. The diagonal terms correspond to either $p_1 = p_3$ and $p_2 = p_4$, or $p_1 = p_4$ and $p_2 = p_3$. They may thus be evaluated directly, approximating the sums by integrals according to the prescription (30), to give

$$\tilde{r}_{\text{diag}}(x_1, x_2, x_3, x_4) \approx \frac{1}{8\pi^4}\int_0^\infty\int_0^\infty \alpha_1\alpha_2 \exp[i(x_1-x_3)\alpha_1 + i(x_2-x_4)\alpha_2]\,d\alpha_1 d\alpha_2 + [x_3 \leftrightarrow x_4] \tag{44}$$

where the second term is obtained by interchanging x_3 and x_4 in the first. Hence

$$\tilde{r}_{\text{diag}}(x_1, x_2, x_3, x_4) \approx \frac{1}{8\pi^4}\left[\frac{1}{(x_1-x_3)^2(x_2-x_4)^2} + \frac{1}{(x_1-x_4)^2(x_2-x_3)^2}\right] \tag{45}$$

and so the contribution to the combination of $\tilde{r}_4^{(2)}$ terms appearing in (36) is

$$\tilde{r}_{\text{diag}}(x_1, x_2, x_3, x_4) + \tilde{r}_{\text{diag}}(x_3, x_2, x_1, x_4) + \tilde{r}_{\text{diag}}(x_4, x_2, x_3, x_1) \approx \frac{1}{4\pi^4}\sum_{i=2}^{4}\frac{1}{(x_1-x_i)^2(x_j-x_{j'})^2} \tag{46}$$

where $j \neq 1, i$; $j' \neq 1, i$; and $j' > j$.

4.2.2. Type-I off-diagonal contributions. The contributions from groups of primes satisfying $p_1p_2 = p_3p_4 + k$ may be calculated as before by replacing one of the prime-sums in (38) by a sum over k, incorporating the associated restrictions via an inclusion/exclusion expansion with respect to the other three primes. Equivalently, and more conveniently in this case, we may sum over all four primes as well as k, imposing the fact that $p_1p_2 \approx p_3p_4$ by the inclusion of a δ-function. In doing so, the restrictions on k to be incorporated in the associated inclusion/exclusion expansion are that $(k, p_i) = 1$, where $i = 1, 2, 3, 4$. This has the benefit of respecting the symmetry of the initial expression.

In this case, making the approximation

$$\exp\left[\mathrm{i}E\ln\left(\frac{p_1p_2}{p_3p_4}\right)\right] \to \exp\left[\frac{\mathrm{i}Ek}{p_3p_4}\right] \tag{47}$$

which follows from the fact that $k \ll p_3p_4$, reduces the k-sum to the general form considered in section 2, and so we can again apply (20) to obtain

$$\begin{aligned}\tilde{r}_I(x_1, x_2, x_3, x_4) \approx -\frac{1}{8\pi^4}\int_0^\infty\int_0^\infty\int_0^\infty\int_0^\infty \delta(\alpha_1+\alpha_2-\alpha_3-\alpha_4)\\ \times \exp(\mathrm{i}x_1\alpha_1 + \mathrm{i}x_2\alpha_2 - \mathrm{i}x_3\alpha_3 - \mathrm{i}x_4\alpha_4)\\ \times \tilde{G}(\alpha_1, \alpha_2, \alpha_3, \alpha_4, E)\,\mathrm{d}\alpha_1\mathrm{d}\alpha_2\mathrm{d}\alpha_3\mathrm{d}\alpha_4\end{aligned} \tag{48}$$

where

$$\begin{aligned}\tilde{G}(\alpha_1, \alpha_2, \alpha_3, \alpha_4, E) = (\ln E - \alpha_3 - \alpha_4)\Theta(\alpha_3 + \alpha_4 - \ln E)\\ -\sum_{i=1}^{4}(\ln E - \alpha_i)\Theta(\alpha_i - \ln E) + (\ln E + \alpha_1 - \alpha_4)\\ \times \Theta(\alpha_4 - \alpha_1 - \ln E) + (\ln E + \alpha_4 - \alpha_1)\Theta(\alpha_1 - \alpha_4 - \ln E)\\ + (\ln E + \alpha_1 - \alpha_3)\Theta(\alpha_3 - \alpha_1 - \ln E)\\ + (\ln E + \alpha_3 - \alpha_1)\Theta(\alpha_1 - \alpha_3 - \ln E)\end{aligned} \tag{49}$$

and the prime-sums have been approximated by integrals using (30). The evaluation of the quadruple integral in (49) is straightforward but tedious. The result is that

$$\begin{aligned}\tilde{r}_I(x_1, x_2, x_3, x_4) \approx \frac{1}{8\pi^4}\,\mathrm{Re}\sum_{i=1}^{2}\sum_{j=3}^{4}\frac{\cos[(x_i - x_j)\ln E]}{(x_i - x_j)^2(x_i - x_{i'})(x_j - x_{j'})}\\ \times\left(1 - \frac{x_i - x_{i'}}{x_j - x_{i'}}\right)\left(1 - \frac{x_j - x_{j'}}{x_i - x_{j'}}\right)\end{aligned} \tag{50}$$

where $i' = 1, 2$, $i' \neq i$, and $j' = 3, 4$, $j' \neq j$. Adding together the combination of $\tilde{r}_4$ terms appearing in (36) then gives

$$\begin{aligned}\tilde{r}_I(x_1, x_2, x_3, x_4) + \tilde{r}_I(x_3, x_2, x_1, x_4) + \tilde{r}_I(x_4, x_2, x_3, x_1)\\ \approx -\frac{1}{4\pi^4}\sum_{i=1}^{4}\sum_{j>i}\frac{\cos[(x_i - x_j)\ln E]}{\prod_{\substack{k\neq i\\ k\neq j}}(x_i - x_k)(x_j - x_k)}\end{aligned} \tag{51}$$

that is, the type-I off-diagonal terms give the same contribution to $\tilde{R}_4$ as the $\tilde{r}_4^{(1)}$ terms.

4.2.3. Semidiagonal contributions. The semidiagonal contributions to $\tilde{r}_4^{(2)}$ involve cases where $p_i = p_j + k$ and $p_{i'} = p_{j'}$, $i \neq i' = 1, 2$; $j \neq j' = 3, 4$. Their evaluation reduces directly to a combination of the methods used to calculate the diagonal terms and the off-diagonal terms (in this case no inclusion/exclusion expansion is necessary, as with the computation of the 2-point correlations in Keating (1993)). The result, obtained using the previous approximations, is that

$$\tilde{r}_{\text{semi}}(x_1, x_2, x_3, x_4) \approx \frac{1}{8\pi^4} \operatorname{Re} \sum_{i=1}^{2} \sum_{j=3}^{4} \frac{\exp[\mathrm{i}(x_i - x_j) \ln E]}{(x_i - x_j)^2} \int_0^\infty \alpha \exp[\mathrm{i}(x_{i'} - x_{j'})\alpha] \, \mathrm{d}\alpha \quad (52)$$

which evaluates to

$$\tilde{r}_{\text{semi}}(x_1, x_2, x_3, x_4) \approx -\frac{1}{8\pi^4} \sum_{i=1}^{2} \sum_{j=3}^{4} \frac{\cos[(x_i - x_j) \ln E]}{(x_i - x_j)^2 (x_{i'} - x_{j'})^2}. \quad (53)$$

Adding together the combination of $\tilde{r}_4^{(2)}$ terms appearing in (36) then gives

$$\begin{aligned} &\tilde{r}_{\text{semi}}(x_1, x_2, x_3, x_4) + \tilde{r}_{\text{semi}}(x_3, x_2, x_1, x_4) \\ &+ \tilde{r}_{\text{semi}}(x_4, x_2, x_3, x_1) \approx -\frac{1}{4\pi^4} \sum_{i=1}^{4} \sum_{j>i} \frac{\cos[(x_i - x_j) \ln E]}{(x_i - x_j)^2 (x_{i'} - x_{j'})^2} \end{aligned} \quad (54)$$

where $i' \neq j' \neq i \neq j$ and $i' > j'$.

4.2.4. Type-II off-diagonal contributions. The fourth and final class of contributions to $\tilde{r}_4^{(2)}$ are those coming from sets of prime-pairings of the form

$$\begin{aligned} mp_1 &= np_3 + k \\ np_2 &= mp_4 + k' \end{aligned} \quad (55)$$

where $(m, n, k) = (m, n, k') = 1$. Such groupings will occur for all integer values of m and n. As in the previous cases, we proceed by replacing the sums over the primes p_3 and p_4 in (38) by sums over the integers k and k', making the replacement

$$\exp\left[\mathrm{i}E \ln\left(\frac{p_1 p_2}{p_3 p_4}\right)\right] \to \exp\left[\mathrm{i}E\left(\frac{k}{mp_1} + \frac{k'}{np_2}\right)\right] \quad (56)$$

(assuming, as will turn out to be case, that the dominant contribution corresponds to when k and k' are small). The appropriate probability factor is again precisely of the form discussed in section 2 and so the k- and k'-sums can be evaluated using the general result (20). The coprimality restrictions mean in this case that the inclusion/exclusion expansion must be performed with respect to the prime factors of both m and n.

The result is that the type-II contributions are given by

$$\begin{aligned} \tilde{r}_{\text{II}}(x_1, x_2, x_3, x_4) \approx \frac{1}{8\pi^4} \operatorname{Re} \int_0^\infty \int_0^\infty \int_0^\infty \int_0^\infty \sum_{(m,n)=1} \frac{1}{mn} &\delta(\alpha_1 + \ln m - \alpha_3 - \ln n) \\ &\times \delta(\alpha_2 + \ln n - \alpha_4 - \ln m) \exp(\mathrm{i}\alpha_1 x_1 + \mathrm{i}\alpha_2 x_2 + \mathrm{i}\alpha_3 x_3 + \mathrm{i}\alpha_4 x_4) \\ &\times (\ln E - \ln m - \alpha_1) \Theta(\alpha_1 + \ln m - \ln E) \\ &\times (\ln E - \ln n - \alpha_2) \Theta(\alpha_2 + \ln n - \ln E) \, \mathrm{d}\alpha_1 \mathrm{d}\alpha_2 \mathrm{d}\alpha_3 \mathrm{d}\alpha_4 \\ &+ \text{inclusion/exclusion} \end{aligned} \quad (57)$$

where the prime-sums have been approximated by integrals according to (30), the δ-functions impose the relations (55), and the sum includes all coprime pairs of integers, corresponding to all possible relations of this type. The inclusion/exclusion series, which has not been written explicitly, is generated by multiplying E by all of the combinations of the prime factors of m and n, p_m and p_n, respectively, the sign of each term being -1 to the power of the number of primes involved. Evaluating the integrals allows this series to be summed, giving

$$\begin{aligned}\tilde{r}_{\mathrm{II}}(x_1,x_2,x_3,x_4) \approx \frac{1}{8\pi^4}\,\mathrm{Re}\sum_{(m,n)=1}\frac{1}{mn}\,\frac{\exp[\mathrm{i}(x_1+x_2-x_3-x_4)\ln E]}{(x_1-x_3)^2(x_2-x_4)^2}\\ \times\exp[\mathrm{i}(x_4-x_1)\ln m]\prod_{p_m}[1-\exp(\mathrm{i}(x_1-x_3)\ln p_m)][1-\exp(\mathrm{i}(x_2-x_4)\ln p_m)]\\ \times\exp[\mathrm{i}(x_3-x_2)\ln n]\prod_{p_n}[1-\exp(\mathrm{i}(x_1-x_3)\ln p_n)][1-\exp(\mathrm{i}(x_2-x_4)\ln p_n)].\end{aligned} \tag{58}$$

To leading order the coprimality restriction may be neglected; that is, m and n may each be summed from 1 to ∞. The expression (58) then factorizes into a product of terms of the form

$$\rho(\alpha,\beta_1,\beta_2)=\sum_{m=1}^{\infty}\frac{1}{m^{1+\mathrm{i}\alpha}}\prod_{p_m}[1-\exp(\mathrm{i}\beta_1\ln p_m)][1-\exp(\mathrm{i}\beta_2\ln p_m)] \tag{59}$$

which, as shown in the appendix, is approximately given by

$$\rho(\alpha,\beta_1,\beta_2)\approx\frac{(\alpha-\beta_1)(\alpha-\beta_2)}{\alpha(\alpha-\beta_1-\beta_2)}. \tag{60}$$

(We note in passing that the derivation given in the appendix justifies the use of (57), which assumes the prime factors of m and n in (55) to be large, because the individual terms in the prime sum (A3) diverge.) Applying this result to (58) leads finally to

$$\tilde{r}_{\mathrm{II}}(x_1,x_2,x_3,x_4)\approx\frac{1}{8\pi^4}\,\frac{(x_1-x_2)^2(x_3-x_4)^2\cos[(x_1+x_2-x_3-x_4)\ln E]}{(x_1-x_3)^2(x_1-x_4)^2(x_2-x_3)^2(x_2-x_4)^2}. \tag{61}$$

In order to generate the total contribution to $\tilde{R}_4$, all of the appropriate permutations of (61) must be added together. First, the combination of $\tilde{r}_4^{(2)}$ terms appearing in (36) gives

$$\begin{aligned}&\tilde{r}_{\mathrm{II}}(x_1,x_2,x_3,x_4)+\tilde{r}_{\mathrm{II}}(x_3,x_2,x_1,x_4)+\tilde{r}_{\mathrm{II}}(x_4,x_2,x_3,x_1)\\ &\quad\approx\frac{1}{8\pi^4}\sum_{i=2}^{4}\frac{(x_1-x_i)^2(x_j-x_{j'})^2\cos[(x_1+x_i-x_j-x_{j'})\ln E]}{(x_1-x_j)^2(x_1-x_{j'})^2(x_i-x_j)^2(x_i-x_{j'})^2}.\end{aligned} \tag{62}$$

where $j>j'\neq 1$, $j\neq i$, $j'\neq i$. Second, it might appear on first sight that one should also include a permutation corresponding to $x_1\leftrightarrow x_2$ in (61) (or, more generally, $x_1\leftrightarrow x_i$ in (62)) related to the interchange of p_1 and p_2 in (55). The effect of this would be to multiply the left-hand side of (62) by an additional factor of 2. It is, however, unnecessary: this prime permutation is already included via the sum over m and n in (58), as may easily be verified by making the substitution

$$\begin{aligned}m&=\tilde{m}p_2\\ n&=\tilde{n}p_1\\ k&=\tilde{k}p_1\\ k'&=\tilde{k}'p_2\end{aligned} \tag{63}$$

in (55). Hence (62) represents the total contribution of the type-II off-diagonal terms to $\tilde{R}_4$.

4.3. The result for $\tilde{R}_4$

Adding together (43), (46), (51), (54) and (62) gives the final approximation for $\tilde{R}_4$. Remarkably, the resulting formula is, as may be checked by a direct but tedious expansion of the determinant, exactly equivalent to the GUE expression (35) if $\bar{d}$ in (24) is given by the mean density of the Riemann zeros (6). This shows that the approximations made are indeed asymptotically consistent, and confirms explicitly that the 4-point correlations of the non-trivial zeros of the zeta function are the same as in the GUE of random matrix theory.

5. Conclusions

We have demonstrated that the 3-point and 4-point irreducible correlation functions of the Riemann zeros are equivalent to the corresponding GUE results of random matrix theory. The methods used were essentially heuristic and involved a number of uncontrolled approximations. Nevertheless, the fact that the final formulae agree exactly with the GUE expressions confirms that they are asymptotically consistent.

Our aim in this paper was to present these calculations in as much detail and as explicitly as possible, so as to focus attention on the essential techniques required to evaluate the various prime-combination contributions. However, as one goes to higher-order correlations, it is not difficult to see that the computations as they stand will become increasingly complicated and rapidly unmanageable. In order to deal with the general nth order case, it is therefore clear that a new combinatorial approach is required to apply these methods effectively. Such an approach will be described in an accompanying publication (Bogomolny and Keating 1995).

One interesting feature of this method of calculation, already apparent for $\tilde{R}_3$ and $\tilde{R}_4$, is that the higher-order correlations of the Riemann zeros can be recovered from the generalized pair-correlation function Π (defined in the first paragraph of section 2) for the primes. In some respects, this is related to the special, arithmetical structure underlying the zeta function; in others, however, it probably represents a more general property, since the irreducible GUE correlation functions $\tilde{R}_n$ are themselves all expressed in terms of the functional form that characterises the 2-point correlations (see equations (3), (23) and (35)). Thus the same behaviour might also be expected for the semiclassical periodic orbit description of the level statistics of classically chaotic, non-time-reversal-symmetric dynamical systems, although the details of how it emerges in that case are, at present, not clear. It is hoped that the investigations reported here will ultimately help to shed some new light on this important issue.

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Appendix

We are to estimate

$$\rho(\alpha, \beta_1, \beta_2) = \sum_{m=1}^{\infty} \frac{1}{m^{1+i\alpha}} \prod_{p_m} [1 - \exp(i\beta_1 \ln p_m)][1 - \exp(i\beta_2 \ln p_m)] \quad \text{(A1)}$$

This may be done as follows. First, the sum can be expressed as a product over all primes p,

$$\rho(\alpha, \beta_1, \beta_2) = \prod_p \left[1 + \frac{(1 - \exp(i\beta_1 \ln p))(1 - \exp(i\beta_2 \ln p))}{p^{1+i\alpha} - 1}\right] \tag{A2}$$

and so the leading-order asymptotics of $\ln \rho$ is that

$$\ln \rho(\alpha, \beta_1, \beta_2) \approx \sum_p \frac{1}{p^{1+i\alpha}} (1 - \exp(i\beta_1 \ln p))(1 - \exp(i\beta_2 \ln p)) . \tag{A3}$$

Approximating the sum by an integral, as in (30) then gives

$$\ln \rho(\alpha, \beta_1, \beta_2) \approx \int_0^\infty \exp(-i\alpha y)(1 - \exp(i\beta_1 y))(1 - \exp(i\beta_2 y)) \frac{dy}{y} \tag{A4}$$

which may be evaluated to yield

$$\ln \rho(\alpha, \beta_1, \beta_2) \approx \ln \left[\frac{(\alpha - \beta_1)(\alpha - \beta_2)}{\alpha(\alpha - \beta_1 - \beta_2)}\right] \tag{A5}$$

or

$$\rho(\alpha, \beta_1, \beta_2) \approx \frac{(\alpha - \beta_1)(\alpha - \beta_2)}{\alpha(\alpha - \beta_1 - \beta_2)} \tag{A6}$$

as stated in the text.

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